hwde-led-test — Design Document

Phase P9; all 5 recorded gates pass.

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Board and Run Metadata

FIELD	VALUE
board	hwde-led-test
part number	none (not in the boards register)
workspace	`/work/boards/hwde-led-test`
phase	P9
gate status	all 5 recorded gates pass
state created	2026-10-28T09:25:11

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FIELD	VALUE
state updated	2026-10-28T10:08:43
design spend	not recorded

Overview

hwde-led-test - brief

The Hive's first hwde board: a tiny, quick test run that proves the pipeline goes from requirements to checked KiCad files inside the Hive. It is not to be ordered; ordering stays the person's.

A tiny 2-layer board that lights one LED from 5 V.

- Input: 5 V DC (4.75-5.25 V) on a 2-pin 2.54 mm pin header, J1: pin 1 = +5V, pin 2 = GND. From a bench supply or a USB 5 V lead. No reverse-polarity protection wanted (the LED itself blocks reverse current; a test board).
- One green LED, 0603, lit at about 3-5 mA through one 0603 series resistor.
- One 0603 ceramic capacitor (1-10 uF, >= 10 V) across the input, at J1.
- Parts from the JLCPCB basic library wherever one fits; SMD parts on the top side only, for JLCPCB assembly; the header may be through-hole.
- Board: 2 layers, 1.6 mm FR-4, HASL, as small as practical (about 15 x 12 mm is fine, no HARD cap), no mounting holes, no enclosure.
- Quantity if ever built: 5. Room temperature, bench use.

Requirements

hwde-led-test - requirements

1. Function

A tiny 2-layer test board that lights one green 0603 LED from a 5 V DC input through one 0603 series resistor, with one 0603 ceramic input capacitor at the input header. Its purpose is to prove the hwde pipeline runs from requirements to checked KiCad files inside the Hive. It is not to be ordered; ordering stays with the person. No declared mode (normal design).

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2. Interfaces

- J1: 2-pin 2.54 mm pin header (through-hole allowed). Pin 1 = +5V, pin 2 = GND. Fed from a bench supply or a USB 5 V lead.
- D1: one green LED, 0603, lit at about 3-5 mA.
- No debug connector, no MCU, no buttons, no other I/O.

3. Power

- Input: 5 V DC, range 4.75-5.25 V, on J1.
- No reverse-polarity protection (stated: the LED blocks reverse current; test board).
- One rail only (+5V). LED current target 3-5 mA via one 0603 series resistor (value to be chosen at design time from the LED Vf; ASSUMED: LED Vf about 2.0-3.0 V for green, so roughly 470 ohm-1 k ohm).
- Input capacitor: one 0603 ceramic, 1-10 uF, >= 10 V, across the input at J1.
- Total load about 5 mA max (guess). No battery, no charging.

4. Environment

Room temperature, bench use. No enclosure, no ingress, no vibration requirement.

5. Size & mounting

- Outline: as small as practical, about 15 x 12 mm is fine. SOFT target, no HARD cap.
- Mounting holes: none. No enclosure.
- Height limit: none stated.
- Stackup: 2 layers, 1.6 mm FR-4, HASL.

6. Quantity & budget

Quantity if ever built: 5. Target unit cost: not stated (ASSUMED: not a constraint).

7. Assembly

JLCPCB PCBA. SMD parts on the top side only (single-sided SMD assembly); the header J1 may be through-hole. Use JLCPCB basic-library parts wherever one fits.

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8. Compliance/safety flags

None apply: no mains, no battery, no motors, no voltage above 30 V, no current above 3 A, no RF transmit. (5 V DC input, about 5 mA load.)

9. Open questions

none

Architecture

Decisions of record (state.json)

PHASE	WHAT	WHY
P1	P1 research roster empty: no scout, reference-design, interface or power researcher	trivially powered (5 V in, one LED branch at ~5 mA); a header, an LED, a resistor and a cap need no research
P2	designing under coverage gap: block:B3 led-indicator (no coverage checklist exists for this topology); research not launched	a single LED + series resistor at 4.3 mA from 5 V is textbook; a research pass would build a checklist for a block with no layout risk. Presented at H1 for the person to overrule
P3	D1 KT-0603G (C12624, Extended, Vf 3.1 V listing value); R1 470R (C23179) not 680R	brief asks for an 0603 green LED and there is no Basic one; at Vf 3.1 V 680R gives 2.4-2.8 mA, below the 3 mA floor, 470R gives 3.5-4.6 mA. Vf is the LCSC listing, the datasheet would not download
P3	hand edit: widened the F.CrtYd of C0603, R0603, LED0603-RD from +-0.80x+-0.40 to +-1.40x+-0.70 mm	the EasyEDA courtyards were the body outline and did not enclose the pads (pads reach +-1.15 x +-0.45); IPC courtyard = pads + 0.25 mm, so placement legality and DRC courtyard checks are not lenient
P3	fp_verify and lib_pin_types not run; D1 cathode = pad 1 from symbol K pin + silk chamfer, not datasheet	no IC on the board, so no datasheet extractions to diff against; the KT-0603G datasheet PDF would not download
P6	outline fit 13.05 x 9.08 mm, margin 2.0, corner r 2.0	margin 1.0/1.5 clipped the J1/R1/D1 refdes silk at the edge; 2.0 keeps all silk on the board and stays under the brief's ~15 x 12 mm
P9	parts.json lines now carry refdes and the LCSC attributes	bom_cpl maps ref->LCSC through refdes; without it BOM.csv had no LCSC column (the schematic reviewer's warning 2)
P10	stop at P9: no order, no quote, no board guide	in the Hive ordering at JLCPCB or any purchase is the person's; this is a pipeline test board, not for ordering

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Sheet plan

Sheet plan

Single sheet: `root` (no hierarchy). No hier pins.

```
| Sheet | Blocks | Nets | Refdes |
|---|---|---|---|
| root | B1 J1, B2 C1, B3 R1+D1 | +5V, GND, /LED_A | J1; C1; R1; D1
(all unique); #PWR range 100+ (pwr_base=100) |
```

Canonical net names: +5V (J1.1, C1.1, R1.1), GND (J1.2, C1.2, D1 cathode), /LED_A (R1.2 - D1 anode).

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

Schematic

Gate `erc` (phase P4): **pass** after 4 attempt(s); last 2026-10-28T10:08:43, failing 0 of 0.

Review waivers

Schematic review warnings carried, not fixed (P4)

- led-current-margin (R1, D1): accepted. The brief asks for ”about 3-5 mA”; at the listed Vf 3.1 V the current is 3.5-4.6 mA. Only an unconfirmed worst-case Vf of 3.4 V at 4.75 V input lands at 2.8 mA, which still lights the LED.
- bom-fields-missing (J1, C1, R1, D1): checked again at P9, where bom_cpl writes the JLC BOM; if the LCSC column is empty there, it is fixed then.

The full schematic PDF follows.

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PWR_FLAG
+5V

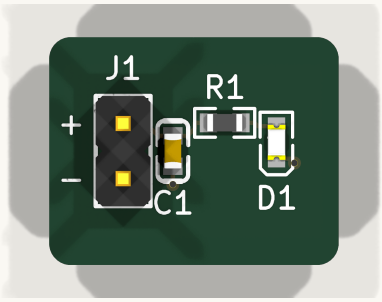
PWR_FLAG
GND

hwde		
Sheet: /		
File: hwde-led-test.kicad_sch		
Title: hwde-led-test: 5 V LED indicator		
Size: A4	Date:	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/1

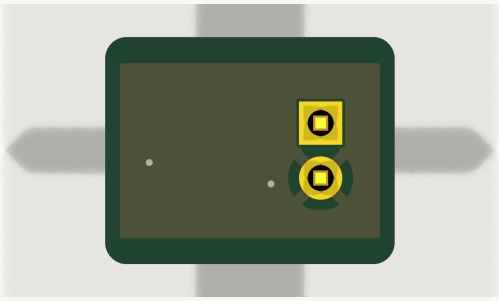
Layout

Gate `place` (phase P6): **pass** after 6 attempt(s); last 2026-10-28T10:05:32, failing 0 of 0.

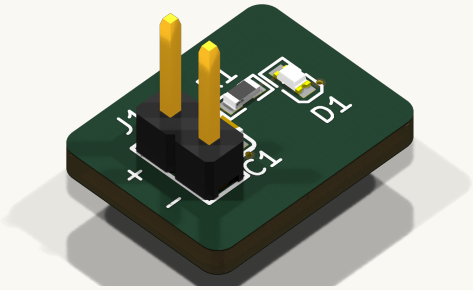
Gate `drc_routed` (phase P7): **pass** after 3 attempt(s); last 2026-10-28T10:05:38, failing 0 of 0.



reports/renders/hwde-led-test_top.png



reports/renders/hwde-led-test_bottom.png



reports/renders/hwde-led-test_iso.png

Verification

Gate `verify` (phase P8): **pass** after 5 attempt(s); last 2026-10-28T10:05:41, failing 0 of 2.

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Verification checks

CHECK	STATUS	FINDINGS	NOTE
check_return_path	pass	0	
check_current	pass	0	
check_decoupling	pass	0	
check_diffpair	pass	0	
check_creepage	pass	0	
check_thermal	pass	0	
check_silk	pass	0	
check_route_style	violations	1	
check_mating	pass	0	
check_pdn	pass	0	
check_ratings	violations	1	
check_bom_sync	pass	0	
check_mate_pins	pass	0	
check_model_paths	pass	0	

Design review of record

hwde-led-test - P8 board review (verify-reviewer, fresh context)

Inputs: reports/checks/summary.json (14/14 checks ran, none skipped, 2 warnings), reports/verify-gate.json (PASS), renders reports/renders/hwde-led-test_top.png, hwde-led-test_bottom.png, hwde-led-test_iso.png (review set, w 2400), reports/hwde-led-test-schematic.pdf, requirements.md, architecture/, kicad/constraints.json, parts/parts.json.

Result: 0 errors, 3 warnings. Nothing found that stops this board working on the bench.

Findings (worst first)

1. WARNING, silk: J1 polarity is not marked (hwde-led-test_top.png, _iso.png). The J1 silk is a plain rectangle with no ”+”, ”5V” or ”GND”. Pin 1’s square pad shows only on the bottom (hwde-led-test_bottom.png) and is hidden under the header’s plastic on top. requirements.md section 3 drops reverse protection because ”the LED blocks reverse current”. With the leads swapped, D1 sees about 5.25 V reverse (C1 and R1 take it with no problem). Green InGaN 0603 LEDs are usually rated VR = 5 V, and parts.json carries no VR for KT-0603G. Marking the pins is the cheap protection. Fix: ”+” / ”-” text at J1 pins 1/2. The bottom side is empty, so B.SilkS fits easily.

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2. WARNING, fab: the board is below JLC's minimum board size for PCBA. The outline is 13.05 x 9.08 mm. reference/jlc_capabilities.yaml records a 10 x 10 mm board minimum, and requirements section 7 says JLCPCB PCBA. The board also has no fiducials. Fix it at the fab step: panelize with rails (JLC puts the fiducials on the rails), or grow the outline to 10 mm or more in Y (placement or outline fit). The P6 decision to fit at 2 mm margin was recorded and section 5 is a soft target, so the size itself is not drift. Only the PCBA minimum is the issue. The board is not to be ordered now, so this is a warning.

3. WARNING, silk: the C1 refdes is ambiguous (hwde-led-test_top.png). The "C1" text sits at about (15.2, 22.3), under the gap between R1 and D1 and roughly equidistant from C1, R1 and D1. Move it beside C1.

Hunt list, cleared

- Connector mating: J1 is a vertical 2.54 mm header with mating from above (iso render). Nothing tall is near it. C1 (0.8 mm tall, pad edge at about 12.7 mm) is clear of a 2.54 mm Dupont housing (header body edge at 12.17). J1 is centred on the left edge as constraints.json asks (edge left, pos 0.5: J1 y 20.2 = board centre).
- Polarity after assembly: the D1 outline has a chamfer at the cathode end, outside the body, so it stays visible after assembly. Footprint pad 1 = K = GND matches symbol pin 1 = K and the schematic. The small diode glyph inside the outline is hidden under the body (not a defect).
- Cross-artifact: all requirements interfaces are present (J1 +5V/GND, D1 green 0603, R1 0603, C1 0603 4.7 uF 16 V X5R at J1). All three architecture blocks are present. No mounting holes, as stated. 2 layers. SMD top side only. The change from R1 680R (architecture) to 470R is a recorded P3 decision (real Vf 3.1 V, giving 3.5-4.6 mA over 4.75-5.25 V, inside the 3-5 mA promise), so it is not drift. Schematic, PCB and parts.json values match. D1 is a JLC extended part. parts.json records that no basic 0603 green exists (only an 0805 one), so that is justified.
- Return path and decoupling: +5V runs J1.1 -> C1.1 -> R1 (the input cap is in the path). C1 GND drops through a via at (13.2, 21.74) to the B.Cu GND pour, about 2.3 mm from J1.2, which is thermal-tied to the pour. D1 cathode has its own via. This is fine at 5 mA.
- EMI, RF, crystal, switcher, thermal: none on this board.
- Routing visual: straight runs and 45-degree runs only. The +5V path steps down into C1's pad and back up to R1, which is deliberate (it routes through the cap pad). It is not a needless detour.

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Warning triage (verify_all)

- check_route_style jog, /LED_A F.Cu (17.06, 18.975), 0.15 mm sidestep: a real but cosmetic defect. R1 pad 2 is at y 18.9 and D1 anode at y 19.05. The cheapest fix is to move D1 from (18, 19.8) to (18, 19.65) so the trace runs straight (placement, then reroute that one net). A waiver is also acceptable for a test board. Recommendation: fix it if anything else is touched, otherwise waive.
- check_ratings rating_unrated D1, R1: RECOMMEND WAIVER. R1 does carry a rating ("Voltage-Supply(Max)" 75 V) that the checker did not recognise. It sees at most 5.25 V. D1 is forward-biased in normal use (about 3.1 V). Its only overstress case is reverse polarity, which is handled by finding 1 and listed under OPEN.

Waivers recommended (human decides at checkpoint 4)

- check_ratings D1/R1 (reason above).
- route_style jog on /LED_A, only if the D1 nudge is not taken.

DFM and Fabrication

Gate `dfm` (phase P9): **pass** after 1 attempt(s); last 2026-10-28T10:07:12, failing 0 of 0. DFM findings by severity: none.

Fabrication outputs

Exported layers: F.Cu, B.Cu, F.Silkscreen, B.Silkscreen, F.Mask, B.Mask, F.Paste, B.Paste, Edge.Cuts. Castellated edges: none (the board has no castellated pads).

Bill of materials

COMMENT	DESIGNATOR	FOOTPRINT	LCSC
4.7uF	C1	C0603	C19666
KT-0603G	D1	LED0603-RD	C12624
PZ254V-11-02P	J1	HDR-TH_2P-P2.54-V-M	C492401
470R	R1	R0603	C23179

0 CPL rotation correction(s) applied; no missing LCSC numbers. *fab/order.json not available.*

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Run Record (Appendix)

Phase digests

P0

- P0 intake: requirements.md written by requirements-analyst, check_requirements clean.
- 5 V on J1 (2-pin header), green 0603 LED + series R at 3-5 mA, 0603 input cap.
- 2L 1.6 mm, ~15x12 mm soft, no holes, JLC basic parts, qty 5.
- No safety flags; OPEN: none.

P1

- P1 research: no researchers spawned (decision recorded): trivially powered, 4 jellybean parts.

P2

- P2 architecture: one sheet, blocks J1 header, C1 4.7 uF/16 V input cap, R1 680R + D1 green 0603 LED (~4.3 mA).
- Stackup JLC2313_1.6, 2L, HASL; size is a P5/P6 output (~15x12 soft).
- Coverage: 1 slot (led-indicator), gap - no checklist; research not launched (decision).
- Cost: parts < $0.30/board.

P3

- P3 parts: J1 PZ254V-11-02P C492401 (Ext), C1 CL10A475KO8NNNC C19666 4.7uF 16V (Basic), R1 470R C23179 (Basic), D1 KT-0603G C12624 green (Ext, Vf 3.1 V listing).
- LED current 3.5-4.6 mA over 4.75-5.25 V. Parts ~$0.22 for 5 boards.
- Library pulled + load-checked 4/4; 0603 courtyards widened by hand (decision); fp_verify not run (no extractions).
- Coverage P3: same single gap (led-indicator), decision recorded.

P4

- P4 schematic: one root sheet from kicad/gen/root.py; ERC gate PASS 0/0; netlist_audit 0.
- Nets +5V {J1.1,C1.1,R1.1}, GND {J1.2,C1.2,D1.K}, /LED_A {R1.2,D1.A}.
- Review (fresh, opus): 0 errors, 2 warnings (LED current margin at unconfirmed Vf max; LCSC field name for BOM) - reports/erc-waivers.md.
- Module fix found here: KiCad 10 symdir libs flattened by bin/hwde for kicad-sch-api.

P5

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- P5 board setup inline: board_init 2L JLC2313_1.6, outline auto 26.2x22.4 mm (provisional, fit after P6), 0 holes, corner r 2.0.
- Self-check: parity 0, setup_violations 0, unconnected 5 (P7's). Fab profile 2layer_1oz, floors 0.127/0.127, via 0.6/0.3.
- rules_gen: capability_class 2layer_1oz, pass.

P6

- P6 placement (opus): manual place_edit, annealing skipped (4 parts). J1 left edge rot 270, C1 beside J1, R1 then D1 to the right.
- HPWL 29.72 -> 14.37 mm; route probe 1.0; place gate PASS 0/0 (re-run after the outline fit).
- Outline fit to 13.05 x 9.08 mm (margin 2.0; smaller margins clipped refdes silk). silk_place clean; DRC only the 5 unrouted.
- Module fix: render.py preflight now finds the AppImage's 3D loaders (bin/hwde shim).

P7

- P7 routing (opus): planes_gen GND pour B.Cu; route_critical skipped (KiCadRoutingTools not installed; +5V at 10 mA needs no critical width); Freerouting rung 1 completed 5 -> 0 unrouted.
- stitch_vias 0 added, plane_repair 0 splits, snap45 0 off-angle, cleanup 0 ops.
- drc_routed gate PASS: 0 errors, 0 warnings, 0 unconnected.

P8

- P8 verify gate PASS (14/14 checks ran): 0 errors, 2 warnings (0.15 mm jog on /LED_A; D1/R1 no recognised voltage-rating attribute).
- First verify attempt FAILED check_pdn (+5V had no cap association): decoupling.json now records C1 against J1.1; re-gated.
- verify-reviewer (fresh, opus): 0 errors, 3 warnings - fixed all 3: outline 13.4 x 10.5 mm (JLC assembly 10 mm min), J1 +/- legend, C1 refdes moved. place/drc_routed/verify re-gated PASS.
- Warnings not waived (a waiver needs the person's approval): asked at H4.

P9

- P9 DFM: fab_export (JLC gerbers + drill, zipped), bom_cpl (BOM/CPL/prebuy, 4/4 LCSC after refdes added to parts.json); dfm gate PASS 0/0.
- Extended parts in prebuy: D1 C12624, J1 C492401.
- Stopped before P10 (ordering is the person's).

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Gate attempt history

TIMESTAMP	GATE	STATUS	FAILING/TOTAL
2026-10-28T09:37:16	erc	pass	0/0
2026-10-28T09:51:03	place	pass	0/0
2026-10-28T09:52:59	place	pass	0/0
2026-10-28T09:54:12	place	pass	0/0
2026-10-28T09:59:09	drc_routed	pass	0/0
2026-10-28T09:59:37	verify	fail	1/3
2026-10-28T10:00:12	erc	pass	0/0
2026-10-28T10:00:13	verify	pass	0/2
2026-10-28T10:00:21	erc	pass	0/0
2026-10-28T10:00:22	drc_routed	pass	0/0
2026-10-28T10:00:25	verify	pass	0/2
2026-10-28T10:01:06	verify	pass	0/2
2026-10-28T10:04:36	place	fail	1/1
2026-10-28T10:04:47	place	pass	0/0
2026-10-28T10:05:32	place	pass	0/0
2026-10-28T10:05:38	drc_routed	pass	0/0
2026-10-28T10:05:41	verify	pass	0/2
2026-10-28T10:07:12	dfm	pass	0/0
2026-10-28T10:08:43	erc	pass	0/0

Issues

(no entries)

Human checkpoints

ID	PHASE	STATUS	WHEN	NOTE
1	P2	not reached		
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

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Generation cost

Not recorded (no reports/cost.json).

Artifact Index

FILE	SIZE	NOTE
state.json	28,159 B	
kicad/hwde-led-test.kicad_pcb	27,269 B	
kicad/hwde-led-test.kicad_sch	28,654 B	
fab/hwde-led-test_gerbers.zip	8,135 B	
fab/BOM.csv	164 B	
fab/CPL.csv	172 B	
brief/brief.md	987 B	
requirements.md	1,948 B	
architecture/blocks.md	916 B	
architecture/constraints.json	730 B	
architecture/power_tree.md	429 B	
architecture/sheets.md	336 B	
architecture/stackup.md	439 B	
reports/board-edit-15x11.json	1,866 B	
reports/board-edit-final.json	1,870 B	
reports/board-edit-fit.json	1,935 B	
reports/bom-cpl.json	4,501 B	
reports/dfm-gate.json	1,721 B	
reports/drc_routed-gate.json	1,066 B	
reports/erc-gate.json	867 B	
reports/erc-waivers.md	436 B	
reports/fab-export.json	2,166 B	
reports/h4-silk-ops.json	522 B	
reports/hwde-led-test-schematic.pdf	48,207 B	
reports/place-drc.json	4,163 B	
reports/place-edit-1.json	573 B	
reports/place-edit-h4.json	504 B	
reports/place-gate.json	1,528 B	
reports/place-layer-views.json	360 B	
reports/place-metrics.json	2,773 B	
reports/place-seed.json	784 B	

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FILE	SIZE	NOTE
reports/review-board.json	1,967 B	
reports/review-board.md	4,716 B	
reports/review-schematic.json	1,128 B	
reports/review-schematic.md	2,966 B	
reports/schematic.pdf	48,207 B	
reports/silk-place.json	1,047 B	
reports/top.net	8,460 B	
reports/verify-gate.json	2,672 B	
reports/verify-summary.json	5,526 B	

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